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Roll No.

TMC-205/MSD-205(B)

M. C. A./M. SC. (DATA SCIENCE) /

M. SC. (PHYSICS)

(SECOND SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

CAREER SKILLS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All questions are compulsory.

(ii) Each question carries 1 mark.

(iii) Use of calculator is not allowed.

Set-B

- 1. In an exam, a candidate got 35% marks and failed by 12 marks. If he had got 48% marks**

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instead, he would have got 14 marks more than the passing marks. Find the passing marks.

- (a) 96
 - (b) 82
 - (c) 200
 - (d) None of the above
2. If the numerator of a fraction is increased by 15% and its denominator is decreased by 8%, the value of the fraction is $\frac{15}{16}$. Find the original fraction.
- (a) $\frac{3}{4}$
 - (b) $\frac{5}{7}$
 - (c) $\frac{4}{5}$
 - (d) $\frac{11}{13}$
3. The price of a commodity increases by 14.28%. By what percentage should the consumption be reduced so that the overall expenditure is constant?
- (a) 14.28%
 - (b) 12.5%
 - (c) 16.66%
 - (d) 20%

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4. How many numbers from 1 to 100 can be expressed as a product of their factors in more than one way?
- (a) 25
 - (b) 75
 - (c) 74
 - (d) 26
5. If 70% of $(x - y) = 30\%$ of $(x + y)$, then what percent of x is y ?
- (a) 20%
 - (b) 50%
 - (c) 40%
 - (d) None of the above
6. The ratio of income and expenditure is 7 : 5. Income increases by 50% and expenditure decreases by 20%. If the initial expenditure is ₹ 15,000, find the final saving.
- (a) ₹ 12,000
 - (b) ₹ 31,500
 - (c) ₹ 19,500
 - (d) ₹ 21,500

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7. Out of the 75 candidates, 12% did not come to the examination center and rest were eligible for the entrance exam. Of those who were eligible to take the test, 9.09% forgot their admit card of the exam. Two-third of those who took the exam passed the exam. 75% of those who passed secured first class in the paper. How many students who pass the exam secured less than first class ?

- (a) 12
- (b) 10
- (c) 20
- (d) 15

8. Yuva gave 25% of a certain amount of money to Ram. From the money Ram received, he spent 20% on buying books and 35% on buying a watch. After the mentioned expenses, Ram has ₹ 2,700 remaining. How much did Yuva have initially ?

- (a) ₹ 16,000

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(b) ₹ 15,000

(c) ₹ 24,000

(d) ₹ 27,000

9. In a cylinder, both height and radius increase by 100%, by what per cent does its volume increase ?

- (a) 700%
- (b) 100%
- (c) 30%
- (d) Can't be determined

10. Which of the following is not a prime number ?

- (a) 137
- (b) 181
- (c) 173
- (d) 183

11. What is the value of M and N respectively if $M39048458N$ is divisible by 8 and 11, where M and N are single digit integers ?

- (a) 7, 8

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(b) 8, 6

(c) 6, 4

(d) 5, 4

12. Which of the following is not a rational number ?

(a) $\frac{2}{3}$

(b) $\frac{5}{7}$

(c) $\frac{7}{9}$

(d) $\sqrt{10}$

13. N is the smallest number greater than 600 which gives a remainder of 5 when divided by both 6 and 9. Find the remainder when N is divided by 5.

(a) 2

(b) 5

(c) 0

(d) None of the above

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14. Find the greatest number, which will divide 260, 281 and 303, leaving 7, 5 and 4 as remainders respectively.

(a) 30

(b) 32

(c) 34

(d) 23

15. What is the LCM and HCF of $\frac{2}{3}$, $\frac{4}{8}$ and $\frac{4}{5}$?

(a) $\frac{1}{30}$, 4

(b) $\frac{1}{60}$, 4

(c) $\frac{1}{30}$, 2

(d) 4, $\frac{1}{30}$

16. What is the least value of x, so that 2×5472 is divisible by 9 ?

(a) 7

(b) 8

(c) 9

(d) 3

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17. What is the remainder when 3^7 is divided by 8?

- (a) 1
- (b) 2
- (c) 3
- (d) 5

18. If $N = 2^3 \times 3^4$, $M = 2^2 \times 3 \times 5$, then find the total number of factors of N that are common with the factors of M.

- (a) 8
- (b) 6
- (c) 18
- (d) 20

19. Find the unit place of $456^{456} \times 234^{234} \times 567^{567} \times 912^{912}$.

- (a) 2
- (b) 4
- (c) 6
- (d) 8

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20. A shopkeeper sells at 20% markup and uses a weight of 800 g instead of 1 kg. Find the profit or loss %.

- (a) 45%
- (b) 50%
- (c) 40%
- (d) 35%

21. On selling mangoes at 36 for a rupee, a shopkeeper loses 10%. How many mangoes should he sell for a rupee in order to gain 8%?

- (a) 28
- (b) 27
- (c) 31
- (d) 30

22. If the cost price of 20 articles is same as the selling price of 25 articles, find the gain or loss percentage.

- (a) 20% gain
- (b) 20% loss
- (c) 30% gain
- (d) 31% gain

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23. The actual price of an article was ₹ 5,000, the successive discount on 10% and 20%, then find the selling price.

- (a) ₹ 2,600
- (b) ₹ 3,600
- (c) ₹ 4,600
- (d) ₹ 5,600

24. In front of a shop a sign said, "Sale of 30% discount + 40% discount". What is the effective discount being offered?

- (a) 58% discount
- (b) 70% discount
- (c) 35% discount
- (d) None of the above

25. Goods are purchased for ₹ 450 and one-third is sold at a loss of 10%. At what profit per cent should the remainder be sold so as to gain 20% on the whole transaction?

- (a) 35%
- (b) 52%
- (c) 47%
- (d) 37%

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26. A milkman mixes water to 125 L of milk to make it 130 L and sells it at a markup of 10%. Find the profit or loss %.

- (a) 14% profit
- (b) 14.4% profit
- (c) 10% profit
- (d) 4% loss

27. A certain sum of money at simple interest amounts to ₹ 1,260 in 2 years and to ₹ 1,350 in 5 years. The rate % per annum is :

- (a) 2.5%
- (b) 3.75%
- (c) 5%
- (d) 7.5%

28. If a certain sum of money becomes four times itself in eight years at S. I. in how many years will it become 28 times itself?

- (a) 56 years
- (b) 50 years
- (c) 72 years
- (d) 80 years

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29. On a sum of money, the simple interest for 2 years is ₹ 320, while the compound interest is ₹ 340, the rate of interest being the same in both the cases. The rate of interest is :

- (a) 15%
- (b) 14.25%
- (c) 12.5%
- (d) 10.5%

30. The difference in C. I. and S. I. for 2 years on a sum of money is ₹.160. If the S. I. for 2 years be ₹ 2,880, the rate percent is :

- (a) $5\frac{5}{9}\%$
- (b) $12\frac{1}{2}\%$
- (c) $11\frac{1}{9}\%$
- (d) 9%

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Directions (Q. Nos. 31-36) : In the sentences given below select one option as your answer out of the four so that the sentence follows parallel structure :

31. When Delores realized that her father had made lima beans and rice for dinner, she _____, claimed to feel nauseous, and excused herself from the table. The bag of stale popcorn in her room would tide her over until breakfast.

- (a) clutched her stomach
- (b) did clutch her stomach
- (c) was clutching her stomach
- (d) None of the above

32. Not only did Hank despise the way June chewed with her mouth open, _____ finding her wet towels all over the bathroom floor.

- (a) but also disliking
- (b) but also disliked
- (c) but he also disliked
- (d) None of the above

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33. To protect her delicate hands, Fran will not rake the yard, do the dishes, or _____.

- (a) scrub the bathtub
- (b) won't scrub the bathtub
- (c) refuse to scrub the bathtub
- (d) None of the above

34. When Diane takes her beagle for a walk, Santana enjoys sniffing for edible garbage, _____, and lunging at squirrels.

- (a) she howls at bicyclists
- (b) howls at bicyclists
- (c) howling at bicyclists
- (d) None of the above

35. After Amanda cashes her paycheck, the money goes to her savings account, cappuccino fund, and _____.

- (a) credit card balance
- (b) paying her credit card bill
- (c) towards her credit card balance
- (d) None of the above

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36. In preparation for her run, Alicia _____, applied sunscreen and increased the volume on her iPod.

- (a) tightened her shoelaces
- (b) was tightening her shoelaces
- (c) did tighten her shoelaces
- (d) None of the above

Directions (Q. Nos. 37-43) : In the following questions, some part of the sentence may have errors. Find out which part of the sentence has an error and select the appropriate option.

37. The Party Chief (a) / and the Chief Minister (b) / expressed his views (c) / on demonetization in India. (d)

38. Mr. Budhrani, (a) / my friend and professor, (b) / is known for (c) / their good behaviour. (d)

39. The manager (a) / I as well as (b) / the clerks (c) / absented themselves yesterday. (d)

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40. Every boy and (a) / every girl (b) / will do his best (c) / to pass the exam. (d)
41. Neither the students (a) / nor the teacher (b) / was in its (c) / class. (d)
42. There is (a) / none who (b) / can help you (c) / in this crucial time. (d)
43. He introduced to (a) / the Chairman as (b) / the host of the (c) / event. (d)

Directions (Q. Nos. 44-50) : In the questions below, rearrange the parts, which are labelled A, B, C, D and E to form a logical order of sentences.

44. (A) The two neighbours never fought each other.
- (B) Fights involving three male fiddler crabs have been recorded, but the status of the participants was unknown.
- (C) They pushed or grappled only with the intruder.

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- (D) We recorded 17 cases in which a resident that was fighting an intruder was joined by an immediate neighbor, an ally.
- (E) We therefore tracked 268 intruder males until we saw them fighting a resident male.
- (a) BEDAC
- (b) DEBAC
- (c) BDCAE
- (d) BCEDA
45. (A) He felt justified in bypassing Congress altogether on a variety of moves.
- (B) At times he was fighting the entire Congress.
- (C) Bush felt he had a mission to restore power to the presidency.
- (D) Bush was not fighting just the democrats.

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(E) Representative democracy is a messy business, and a CEO of the White House does not like a legislature of second guessers and time wasters.

- (a) CAEDB
- (b) DBAEC
- (c) CEADB
- (d) ECDBA

46. (A) In the west, Allied Forces had fought their way through southern Italy as far as Rome.

(B) In June 1944 Germany's military position in World War II appeared hopeless.

(C) In Britain, the task of amassing the men and materials for the liberation of northern Europe had been completed.

(D) The Red Army was poised to drive the Nazis back through Poland.

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(E) The situation on the eastern front was catastrophic.

- (a) EDACB
- (b) BEDAC
- (c) BDECA
- (d) CEDAB

47. (A) Experts such as Larry Burns, head of research at GM, reckon that only such a full hearted leap will allow the world to cope with the mass motorisation that will one day come to China or India.

(B) But once hydrogen is being produced from biomass or extracted from underground coal or made from water, using nuclear or renewable electricity, the way will be open for a huge reduction in carbon emissions from the whole system.

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(C) In theory, once all the bugs have been sorted out, fuel cells should deliver better total fuel economy than any existing engines.

(D) That is twice as good as the internal combustion engine, but only five percentage points better than a diesel hybrid.

(E) Allowing for the resources needed to extract hydrogen from hydrocarbon, oil, coal or gas, the fuel cell has an efficiency of 30%.

(a) CEDBA

(b) CEBDA

(c) AEDBC

(d) ACEBD

48. (A) But this does not mean that death was the Egyptians' only pre-occupation.

(B) Even Papyri come mainly from pyramid temples.

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(C) Most of our traditional sources of information about the old Kingdom are monuments of the rich like pyramids and tombs.

(D) Houses in which ordinary Egyptians lived have not been preserved, and when most people died they were buried in simple graves.

(E) We know infinitely more about the wealthy people of Egypt than we do about the ordinary people, as most monuments were made for the rich.

(a) CDBEA

(b) ECDAB

(c) EDCBA

(d) DECAB

49. (A) Too much of the labour movement, it symbolises the brutality of the upper classes.

(B) And to everybody watching, the current mess over foxhunting symbolises the government's weakness.

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(C) To foxhunting's supporters, Labour's 1991 manifesto commitment to ban it symbolises the party's metropolitan roots and hostility to the countryside.

(D) Small issues sometimes have large symbolic power.

(E) To those who enjoy thundering across the countryside in red coats after foxes, foxhunting symbolises the ancient roots of rural lives.

(a) DEACB

(b) ECDBA

(c) CEADB

(d) DBAEC

50. (A) In the case of King Merolchazzar's courtship of the Princess of the Outer Isles, there occurs a regrettable hitch.

(B) She acknowledges the gifts, but no word of a meeting date follows.

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(C) The monarch, hearing good reports of a neighbouring princess, dispatches messengers with gifts to her court, beseeching an interview.

(D) The princess names a date, and a formal meeting takes place; after that everything buzzes along pretty smoothly.

(E) Royal love affairs in olden days were conducted on the correspondence method.

(a) ACBDE

(b) ABCDE

(c) ECDAB

(d) ECBAD

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